Machine Learning Question Bank

Concise Reference Solutions Guide

UNIT III: MODEL AND FEATURE SELECTION (SUPERVISED LEARNING)

• Define regression.

Regression is a type of supervised learning task where the target goal is to predict a continuous, numeric output variable based on one or more independent input features.

• Explain linear regression with a suitable example.

Linear Regression models the relationship between a dependent variable and independent variables by fitting a linear equation (a straight line) to the observed data points. The formula is $Y = mX + c$, where m is the slope and c is the intercept.

> **Example:** Predicting House Prices based on their floor area (square feet). As the square footage increases, the price increases linearly.

• Explain different types of linear regression.

- **Simple Linear Regression:** Involves exactly one independent input variable to predict a single continuous output (e.g., predicting weight based solely on height).
- **Multiple Linear Regression:** Involves two or more independent input variables to predict a single continuous output (e.g., predicting house prices using area, number of bedrooms, and age of the property).

• Explain polynomial regression with a suitable example.

Polynomial Regression is a form of regression analysis in which the relationship between the independent variable X and the dependent variable Y is modeled as an n-th degree polynomial. It fits a curved line to non-linear data points using the equation $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \dots + \beta_n X^n$.

> **Example:** Modeling the growth rate of an epidemic over time. Growth is initially slow, accelerates rapidly, and eventually flattens out, which cannot be captured by a straight line.

- **Explain different types of polynomial regression.**

Polynomial regression is generally classified by the degree of the polynomial equation applied to the features:

- **Linear Degree (1st Degree):** Becomes equivalent to standard linear regression ($Y = \beta_0 + \beta_1 X$).
- **Quadratic Regression (2nd Degree):** Fits a single parabolic curve to the data points ($Y = \beta_0 + \beta_1 X + \beta_2 X^2$).
- **Cubic Regression (3rd Degree):** Contains two turning bends, fitting more complex directional variations ($Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3$).

- **Explain overfitting and underfitting.**

- **Underfitting:** Occurs when a model is too simple to capture the underlying pattern of data. It performs poorly on both training and testing datasets (High Bias, Low Variance).
- **Overfitting:** Occurs when a model learns the training data, including noise and outliers, too perfectly. It performs excellently on training data but fails to generalize to new, unseen testing data (Low Bias, High Variance).

- **Explain logistic regression with a suitable example.**

Logistic Regression is a supervised learning algorithm used to predict categorical probability outcomes (usually binary classification like 0 or 1). It applies a Sigmoid activation function to map any real-valued number into a strict range between 0 and 1.

> **Example:** Predicting whether an email is "Spam" (1) or "Not Spam" (0) based on keywords and character frequencies.

- **Explain different types of logistic regression.**

- **Binary Logistic Regression:** The target categorical output has only two possible outcomes (e.g., Pass/Fail, Yes/No).
- **Multinomial Logistic Regression:** The target output has three or more unordered categorical outcomes (e.g., predicting preferred travel choices: Car, Bike, or Train).
- **Ordinal Logistic Regression:** The target output has three or more ordered categorical classes (e.g., Customer satisfaction rating: Poor, Average, Excellent).

- **Explain KNN algorithm with a suitable example.**

K-Nearest Neighbors (KNN) is a non-parametric, lazy learning algorithm. To classify an unseen query data point, it computes the distance (e.g., Euclidean distance) between that query point and all training data points, selects the K closest neighbors, and assigns the class by a majority vote.

> **Example:** Classifying a newly added online item as "Premium" or "Basic". If $K=3$, and 2 of its closest neighboring items are labeled Premium, the new item is classified as Premium.

• Explain basic components of Bayes theorem.

Bayes' Theorem calculates conditional probability using the mathematical formula:

$$P(A|B) = [\, P(B|A) \times P(A) \,] / P(B)$$

The basic components are:

- $P(A|B)$ **(Posterior Probability):** The probability of hypothesis A occurring given that event B has occurred.
- $P(B|A)$ **(Likelihood):** The probability of observing event B given that hypothesis A is true.
- $P(A)$ **(Prior Probability):** The initial probability of hypothesis A before looking at any evidence.
- $P(B)$ **(Marginal Likelihood / Evidence):** The total baseline probability of event B occurring.

• Explain Naive Bayes theorem with a suitable example.

Naive Bayes is a classification algorithm based on Bayes' Theorem. It is called "Naive" because it makes the strong, simplified assumption that all input features are entirely independent of one another.

> **Example:** Weather classification to decide if one should play sports based on Outlook (Rainy/Sunny) and Temperature. It evaluates the independent probabilities of each condition to determine the final probability of "Play" vs "Don't Play".

• Explain Decision Tree algorithm with a suitable example.

A Decision Tree is a flowchart-like tree structure where internal nodes represent tests on features, branches represent the decision outcomes, and leaf nodes represent the final class labels or continuous predictions. Data is split iteratively based on information gain or Gini impurity metrics.

> **Example:** Loan approval systems. Root node: Credit Score > 700? If Yes → Approve. If No → Check Income level → If Low → Reject, If High → Approve.

• Explain Random Forest algorithm with a suitable example.

Random Forest is an ensemble learning method that builds a large collection of independent decision trees during training. It uses a technique called bagging (bootstrap aggregating) to train individual trees on different subsets of data. The final prediction is determined by a majority vote for classification or taking the average for regression.

> **Example:** Recommending a movie. Instead of asking one single expert (one decision tree), you ask 100 diverse people (Random Forest) and pick the movie that gets the most votes.

• Outline working of Support Vector Machine (SVM) with its types.

SVM works by finding an optimal hyperplane in an N-dimensional space that maximizes the geometric margin distance between distinct data classes. The data points nearest to the decision hyperplane boundary are called support vectors.

Main Types:

- **Linear SVM:** Used for linearly separable data points; a basic straight line or flat hyperplane splits the boundaries cleanly.
- **Non-Linear SVM:** Used when data cannot be cleanly split by a straight line. It uses a "Kernel Trick" (e.g., RBF, Polynomial) to project data into higher dimensions where it becomes linearly separable.

• Explain bias and variance with its types.

- **Bias:** Error introduced by simplifying assumptions made by a model to make the target function easier to learn. High bias leads to *underfitting* because the model ignores critical relationships.
- **Variance:** The amount that the target model's estimation function will change if a entirely different training dataset is substituted. High variance leads to *overfitting* because the model is hyper-sensitive to training noise.

UNIT IV: UNSUPERVISED LEARNING

• Define clustering. Explain different types of clustering.

Clustering is the unsupervised learning process of partitioning a set of unlabeled data points into distinct groups (clusters) so that items within the same cluster are highly similar, while items across different clusters are highly distinct.

Main Types:

- **Partitioning Methods:** Divides data directly into non-overlapping groups (e.g., K-Means).
- **Hierarchical Methods:** Creates a multi-level nested tree structure of clusters (e.g., Agglomerative HCA).
- **Density-Based Methods:** Connects spatial dense regions separated by sparse noise zones (e.g., DBSCAN).

• Explain K-Means Clustering algorithm with a suitable example.

K-Means partitions data into K distinct clusters. **Working Steps:** (1) Randomly place K cluster centroids. (2) Assign each data point to its nearest centroid. (3) Recalculate centroids by averaging the coordinates of all points assigned to that cluster. (4) Repeat steps 2-3 until centroids stop shifting.

> **Example:** Segmenting e-commerce customers into 3 clusters based on age and annual spending scores to run targeted marketing ads.

• Explain HCA Clustering algorithm with a suitable example / Outline working of HCA.

Hierarchical Cluster Analysis (HCA) builds a tree-like hierarchy of clusters called a **Dendrogram**. The most common type is Agglomerative (Bottom-Up).

Working Steps: (1) Treat every single data point as an individual cluster. (2) Compute a distance matrix and merge the two closest clusters. (3) Recalculate distances between groups. (4) Repeat merging until all data points are combined into one single root cluster.

> **Example:** Grouping animal species into an evolutionary family tree based on shared genetic similarity metrics.

• Explain DBSCAN algorithm with a suitable example.

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) groups points based on spatial density. It requires two parameters: **Epsilon (eps)** (search radius) and **MinSamples** (minimum points to form a dense region). Points are categorized as Core, Border, or Noise/Outliers. It easily identifies arbitrary shapes and filters noise.

> **Example:** Identifying geographical clusters of crime zones in a city while ignoring isolated random incidents across a map.

• Define dimensionality reduction.

Dimensionality Reduction is the process of reducing the total number of input features or random variables under consideration in a dataset, by obtaining a set of principal or principal-like variables while retaining critical dataset information.

• Explain working of PCA.

Principal Component Analysis (PCA) is an unsupervised technique that projects high-dimensional data onto orthogonal axes representing maximum variance.

Working Steps: (1) Standardize the feature dataset. (2) Compute the covariance matrix. (3) Calculate eigenvectors and eigenvalues from it. (4) Sort eigenvalues in descending order to identify top Principal Components. (5) Project the original data points onto these new axes.

• Explain working of LDA.

Linear Discriminant Analysis (LDA) is a supervised dimensionality reduction technique used to maximize class separability.

Working Steps: (1) Compute the within-class scatter matrix (S_W) and between-class scatter matrix (S_B). (2) Calculate the eigenvectors and eigenvalues for the matrix product $S_W^{-1}S_B$. (3) Select the top eigenvectors to create a lower-dimensional linear subspace that maximizes class distance while minimizing internal variance.

• Define anomaly detection.

Anomaly Detection is the task of identifying rare items, unexpected events, or observations in data that deviate significantly from the established normal patterns of the dataset.

• Explain Isolation Forest algorithm with a suitable example.

Isolation Forest explicitly isolates anomalies instead of profiling normal data. It constructs multiple random decision trees. Because anomalies possess distinct characteristics and are sparse, they require significantly fewer random splits to isolate and appear much closer to the root node of trees (shorter path lengths).

> **Example:** Identifying credit card fraud. A massive fraudulent transaction made overseas isolates quickly on a random split compared to thousands of typical local grocery purchases.

• Explain IQR method of Isolation Forest algorithm / outlier detection with a suitable example.

The Interquartile Range (IQR) method is a statistical technique used to detect outliers.
Working Steps: (1) Sort data and find $Q1$ (25th percentile) and $Q3$ (75th percentile). (2) Calculate $IQR = Q3 - Q1$. (3) Set upper/lower boundaries: $Lower = Q1 - 1.5 \times IQR$ and $Upper = Q3 + 1.5 \times IQR$. Points falling outside this range are flagged as outliers.

> **Example:** Checking salary distributions. If $Q1=30k$, $Q3=70k$, $IQR=40k$. Limits are -30k and 130k. An executive earning 500k falls well above 130k and is flagged as an outlier.

• Explain Z-Score method of Isolation Forest algorithm / outlier detection with a suitable example.

The Z-Score method identifies outliers by measuring how many standard deviations a specific data point lies away from the population mean. Formula: $Z = (X - \mu) / \sigma$. In a standard normal distribution, data points with a absolute Z-score value greater than 3 ($|Z| > 3$) are flagged as outliers.

> **Example:** Quality control testing of bolt lengths. If the average bolt length is 10cm (μ) with a standard deviation (σ) of 0.1cm, a defective bolt measuring 10.5cm yields a Z-score of +5, flagging it instantly as an anomaly.

UNIT V: MODEL OPTIMIZATION TECHNIQUES

• Define optimization.

Optimization in Machine Learning is the iterative process of adjusting a model's internal hyperparameters or weights to minimize a given error/loss function, thereby maximizing the model's accuracy on predictive metrics.

• Explain gradient descent with a suitable example.

Gradient Descent is an optimization algorithm used to minimize a model's loss function. It calculates the partial derivative gradient of the loss function at the current position and updates parameters in the opposite direction of the steepest ascent (downwards) using a step parameter called the **Learning Rate (α)**.

> **Example:** Imagine being blindfolded near the top of a foggy mountain bowl. To reach the bottom lake (global minimum), you feel the slope of the ground with your feet and repeatedly take small steps downhill in the direction of the steepest downward incline.

• List different ensemble methods and explain each in detail.

Ensemble methods combine multiple base models to produce a more robust, generalized prediction.

- **Bagging (Bootstrap Aggregating):** Trains multiple base models in parallel on independent random subsets of training data sampled with replacement. Final outputs are averaged or voted on (e.g., Random Forest). Reduces variance.
- **Boosting:** Trains base models sequentially. Each new model focuses explicitly on correcting the classification mistakes or errors made by its immediate predecessor by adjusting training data weights (e.g., AdaBoost, Gradient Boosting). Reduces bias.
- **Stacking:** Trains completely diverse types of base models (e.g., SVM, Decision Tree, KNN) in parallel, and feeds their individual predictions as meta-features into a final structural meta-model to output the ultimate prediction.

• Define reinforcement learning.

Reinforcement Learning (RL) is a feedback-driven branch of machine learning where an autonomous **Agent** learns to navigate an unfamiliar **Environment** by executing actions and receiving feedback in the form of numerical **Rewards** or punishments.

• Explain working of MDP with a suitable example.

Markov Decision Process (MDP) provides a mathematical framework for modeling decision-making in reinforcement learning environments. It is defined by a 5-tuple: **(S, A, P, R, γ)** where S is States, A is Actions, P is Transition Probability, R is Reward, and γ is the Discount Factor. It operates on the *Markov Property*: the future state depends exclusively on the current state and action, completely independent of past history.

> **Example:** A grid-world chess or navigation game. The robot is at state coordinates (0,1). It chooses action 'Move Right'. The environment transitions it to new state (1,1) and gives a reward of +10 points for avoiding an obstacle.

UNIT VI: ARTIFICIAL NEURAL NETWORK AND DEEP LEARNING

• Write a short note on ANN.

An Artificial Neural Network (ANN) is an information-processing computational paradigm inspired by the structural architecture of the biological human brain. It consists of interconnected artificial processing elements called nodes or neurons, arranged across distinct functional layers. ANNs learn complex non-linear mappings by adjusting interconnection weights during training.

- **Draw and explain the architecture of ANN.**

The fundamental architecture of an ANN consists of three primary layered components:

- **Input Layer:** Passive nodes that receive raw independent feature vectors from external sources and pass them forward without computation.
- **Hidden Layer(s):** Intermediate processing units where neurons apply internal interconnection weights (W), sum up signals, add a baseline bias (b), and pass the result through a non-linear activation function (e.g., ReLU, Sigmoid) to extract features.
- **Output Layer:** Renders the final predicted classification or continuous scalar outcome of the target network.

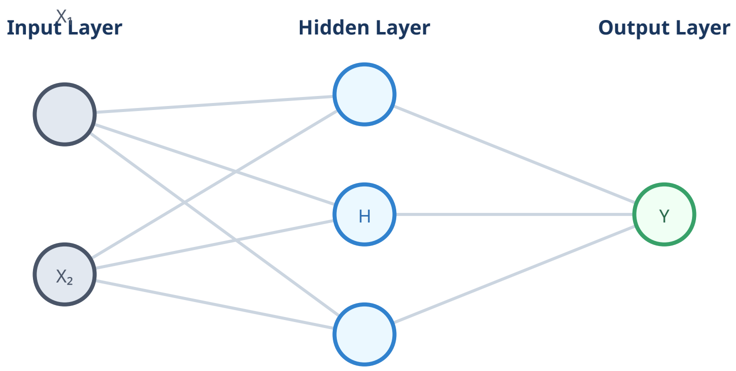

Figure 1: Multilayer Feedforward Neural Network Architecture

- **Explain working of ANN.**

An ANN operates through two main functional data tracking cycles:

1. **Forward Propagation:** Input values enter the input layer, get scaled by connection weights (W), and add a bias value (b). Neurons evaluate activation functions to generate intermediate outputs, passing them onward to the next layer until a final prediction is generated at the output layer.
2. **Loss Evaluation:** The final output is compared against the actual target ground truth label using a loss function to calculate error rates.
3. **Weight Adaptation:** Optimization loops back to tune internal parameters to lower this error over time.

- **Outline working of backpropagation with a suitable example.**

Backpropagation is the core algorithm used to train neural networks by updating weights via the **Chain Rule** of calculus.
Working Sequence: (1) Run forward propagation to obtain a prediction. (2) Compute the network's prediction error at the output layer. (3) Propagate this error gradient backwards through the hidden layers. (4) Compute the partial derivatives of the error with respect to each individual weight. (5) Use Gradient Descent to update all system weights to minimize loss.

> **Example:** Image classification task. The network sees a picture of a "Cat" but incorrectly predicts "Dog" with 80% certainty. The system computes the error delta, flows the gradient backward through layers, and lowers weights connected to "dog" features while strengthening "cat" feature paths.

• Define deep learning and explain different types of neural networks.

Deep Learning is a specialized subfield of Machine Learning based on Artificial Neural Networks that feature multiple (deep) hidden processing layers to automatically discover and extract feature hierarchies from raw unstructured data.

Key Types of Neural Networks:

- **Feedforward Neural Network (FNN):** The simplest type of ANN, where information moves strictly forward in one single direction from inputs to outputs without any recursive loops.
- **Convolutional Neural Network (CNN):** Features convolutional filter matrices to slide over data grids. Optimized heavily for processing spatial data patterns like images and videos.
- **Recurrent Neural Network (RNN):** Contains internal feedback memory loops allowing information to persist across sequential steps. Ideal for sequential or time-series data streams like text and audio.